# Joshua N. **Grant**

*Geospatial Systems Architect*

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## Objective

Transform theories, ideas, and research into production-scale geospatial, cloud, and data solutions. Passionate about scalable architectures, machine learning, and automation.

## Skills

**Languages:** Python, R, JavaScript, SQL, Bash, LaTeX
**Databases:** PostgreSQL, MongoDB, MySQL, Snowflake
**Visualization:** Shiny, Leaflet, Kibana, Plotly
**Cloud:** AWS, GCP, Azure
**GIS:** ArcGIS, GeoPandas, PostGIS, GDAL, QGIS
**ML/AI:** TensorFlow, Scikit-learn, NLP, AI Deployment
**Data Eng.:** Airflow, Kafka, Prefect, Hadoop
**DevOps:** Docker, Kubernetes, Ansible, GitHub Actions
**Version Control:** Git, GitHub, GitLab

## Experience

**Geospatial Systems Architect** *Oak Ridge National Lab* 2023–Present
Architected cloud-native geospatial pipelines, designed a geoparquet-based data warehouse, and developed real-time parcel monitoring with ML models.

**Senior Software Engineer – Data Engineering** *Bold Penguin (Remote)* 2022–2023
Migrated workflows to Prefect 2, built AWS Fargate microservices, and automated data ingestion pipelines.

**Data Engineer** *Oak Ridge National Lab* 2018–2022
Led large-scale raster ETL pipeline development, built misinformation classification models, and optimized geospatial processing.

**Bioinformatician** *Microbial Insights* 2016-2018
Managed projects, developed bioinformatics pipelines, and provided technical support for environmental DNA analysis.

## Projects

**This Is A Casino**
A machine-learning-driven platform for predicting and trading stocks using semantic learning and reinforcement learning. **Technologies:** Python, Flask, TensorFlow, Scikit-learn, Docker, PostgreSQL, React.

**Where I've Been**
A web application visualizing user travel history with dynamic map interactions and real-time database updates. **Technologies:** Python, FastAPI, PostgreSQL, Leaflet, Vue.js, Bootstrap, Docker.

## Education

M.S. Plant Sciences – Molecular GeneticsUniversity of Tennessee, 2017
B.S. Plant Sciences – BiotechnologyUniversity of Tennessee, 2014